

MARK SCHEME for the May/June 2010 question paper
for the guidance of teachers

9702 PHYSICS

9702/23

Paper 2 (AS Structured Questions), maximum raw mark 60

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the May/June 2010 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2010	9702	23

- 1 (a) (i) 1% of ± 2.05 is ± 0.02 A1 [1]
(ii) max. value is 2.08 V A1 [1]
(b) there may be a zero error/calibration error/systematic error M1
which makes all readings either higher or lower than true value A1 [2]
- 2 (a) no resultant force/sum of forces zero B1
no resultant moment/torque/sum of moments/torques zero B1 [2]
(b) (i) each force is represented by the side of a triangle/by an arrow M1
in magnitude and direction A1
arrows joined, head to tail B1 [3]
(*could be shown on a sketch diagram*)
(ii) if the triangle is 'closed' (then the forces are in equilibrium) B1 [1]
(c) triangle drawn with correct shape (incorrect arrows loses this mark) B1
 $T_1 = 5.4 \pm 0.2 \text{ N}$ B1
 $T_2 = 4.0 \pm 0.2 \text{ N}$ B1 [3]
(d) forces in strings would be horizontal B1
(so) no vertical force to support the weight B1 [2]
- 3 (a) evidence of use of area below the line B1
distance = 39 m (*allow $\pm 0.5 \text{ m}$*) A2 [3]
(*if $> \pm 0.5 \text{ m}$ but $\leq 1.0 \text{ m}$, then allow 1 mark*)
(b) (i) 1 $E_K = \frac{1}{2}mv^2$ C1
 $\Delta E_K = \frac{1}{2} \times 92 \times (6^2 - 3^2)$
= 1240 J A1 [2]
2 $E_P = mgh$ C1
 $\Delta E_P = 92 \times 9.8 \times 1.3$
= 1170 J A1 [2]
(ii) $E = Pt$ C1
 $E = 75 \times 8$
= 600 J A1 [2]
(c) (i) energy = $(1240 + 600) - 1170$ M1
= 670 J A0 [1]
(ii) force = $670/39 = 17 \text{ N}$ A1 [1]
(d) frictional forces include air resistance B1
air resistance decreases with decrease of speed B1 [2]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2010	9702	23

- 4 (a) (i) solid has fixed volume and fixed shape/incompressible B1 [1]
(ii) gas fills any space into which it is put B1 [1]
- (b) atoms/molecules have (elastic) collisions with the walls (of the vessel) B1
momentum of atom/molecule changes B1
so impulse (on wall)/force on wall B1
random motion/many collisions (per unit time) gives rise to
(constant) force/pressure B1 [4]
- (c) spacing (much) greater in gases than in liquids/about ten times C1
either spacing depends on $1/\sqrt[3]{\rho}$
or ratio of spacings is about 8.8 A1 [2]
- 5 (a) (i) 1 number of oscillations per unit time (not per second) B1 [1]
2 $n\lambda$ A1 [1]
- (ii) $v = \text{distance} / \text{time} = n\lambda/t$ M1
 $n/t = f$ hence $v = f\lambda$ A1
or f oscillations per unit time so $f\lambda$ is distance per unit time M1
distance per unit time is v so $v = f\lambda$ A1 [2]
- (b) (i) 1.0 period is $3 \times 2 = 6.0$ ms C1
frequency = $1/(6 \times 10^{-3}) = 170$ Hz A1 [2]
- (ii) wave (with approx. same amplitude and) with correct phase difference B1 [1]
- 6 (a) (i) movement/flow of charged particles B1 [1]
(ii) work done per unit charge (transferred) B1 [1]
- (b) straight line through origin B1
resistance = V/I , with values for V and I shown M1
= 20Ω A0 [2]
(using the gradient loses the last mark)
- (c) (i) 0.5 A A1 [1]
- (ii) *either* resistance of each resistor is 20Ω *or* total current = 0.8 A C1
either combined resistance = 10Ω *or* $R = E/I = 10 \Omega$ A1 [2]
- (d) (i) 10 V A1 [1]
- (ii) power = EI C1
= $10 \times 0.2 = 2.0 \text{ W}$ A1 [2]

Page 4	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2010	9702	23

- 7 (a) (i) *either* helium nucleus
or particle containing two protons and two neutrons B1 [1]
- (ii) allow any value between 1 cm and 10 cm B1 [1]
- (b) (i) energy = $(8.5 \times 10^{-13}) / (1.6 \times 10^{-13})$
= 5.3 MeV M1
A0 [1]
- (ii) number = $(5.3 \times 10^6) / 31$
= 1.7×10^5 (allow 2 s.f. only) C1
A1 [2]
- (iii) number per unit length = $(1.7 \times 10^5) / \text{(a)(ii)}$
correct numerical value A1
correct unit B1 [2]